William S. Parker

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SUMMARY

Experimental physicist and R&D engineer with a background in materials science, electron optics, laser optics, and advanced technology development. I focus on turning first-principles physics into functional technology, having led projects from pathfinding research through to hands-on-hardware execution.

EDUCATION

Ph.D., Physics — University of Oregon
Awarded: *OMQ Director's Fellowship, Graduate First Year Fellowship* — Dec. 2024

B.S., Double Major: Physics and Mathematics — Chapman University
Awarded: *Outstanding Senior in Physics, Magna Cum Laude* — May 2018

EXPERIENCE

Research Physicist — May 2025 – present
AstroForge, Inc., Seal Beach, CA

- Lead the R&D of asteroid mining processes, setting research direction and growing the team from one person to a cross-functional group of nine.
- Established core process physics, based on laser interactions with ferrous materials and PGMs. Combined test campaigns and analytical modeling to retire alternative solutions and set design baselines.
- Took lab-scale prototypes to hardened, automated systems, validating mining technology in a terrestrial mine.
- Designed and integrated high-power fiber, high-energy pulsed, and ultrafast optical trains with precision motion control and metrology; engineered packaging for dust, contamination, vibration, and thermal cycling.
- Built the full lab hardware/software stack: industrial controls electronics, microcontroller I/O boards, sensor/instrument selection, C++ and Python for control loops, automation, and logging.
- Rapidly prototype optical/optomechanical payloads. CAD, 3D print, machine, and iterate custom assemblies. Build circuitry & code to integrate sensors, power supplies, actuators, and microcontrollers.
- Use, manage, and maintain laboratory hardware: high-energy pulsed lasers, high-power fiber lasers, optical trains, vacuum systems, electron microscope, optical microscopes, induction & muffle furnaces, metrology devices.

PhD Candidate — Mar. 2020 – Dec. 2024
University of Oregon, Eugene, OR

- Showed that hybrid skyrmions in magnetic multilayers are half-integer hopfions, developing a new topological framework for these textures (first-author, *Physical Review B*, 2024).
- Experimentally measured the 3D structure of magnetic skyrmions by combining novel electron microscopy techniques with high-performance numerical modeling.
- Developed holographic techniques for atomic-resolution magnetic imaging at Lawrence Berkeley National Lab; designed and fabricated micromachined electron-optical phase elements for diffractive two-beam interferometry.
- Built and published open-source Python tools for simulating, analyzing, and visualizing electron optics data (Lorentz TEM, electron interferometry).
- Brought custom SEM hardware online, including a custom electron polarization analysis system.
- Initiated and led cross-institutional collaborations with LLNL-NCEM, UC San Diego, and RIKEN.
- Presented results at 6 international scientific conferences, initiating industry and academic collaborations.
- Wrote and published first-author works in Physical Review B and Microscopy & Microanalysis.

Graduate Research Assistant — Jun. 2018 – Mar. 2020
University of Oregon, Eugene, OR

- Created orbital angular momentum electron modes, by fabricating, testing, and modeling diffractive electron optics elements using a focused ion beam microscope.
- Numerically modeled a free-space orbital angular momentum sorter using finite-element and Fourier optics.

Undergraduate Researcher Feb. 2017 – May 2018

Chapman University, Orange, CA

- Generated custom optical vortex modes using a digital micromirror device for Raman spectroscopy applications.

Undergraduate Researcher (National Science Foundation REU, Joseph Eberly) May 2017 – Jun. 2018

University of Rochester, Rochester, NY

- Numerically investigated fundamental limitations and potential improvements to mode-locked ultrafast lasers.

SKILLS

- **Optics & Microscopy:** SEM/FIB, (S)TEM, magnetic imaging, optical mode shaping, (electron) interferometry, spectroscopy (EELS, EDS, Auger, X-ray, IR), high-power/pulsed/ultrafast laser optics
- **Engineering & Prototyping:** Rapid prototyping (3D printing, metal machining), mechanical design (CAD/Fusion), opto-mechanics, PCB design, analog/digital electronics, system testing
- **Computation & Simulation:** Micromagnetics modeling, charged particle optics, signal/image processing (OpenCV, FIJI, custom), finite-element analysis, machine learning
- **Controls & Integration:** Industrial automation, motion control, data acquisition, hardware/software integration, microcontroller I/O, ultra-high vacuum (UHV) systems
- **Software:** Python, JS/Node/Svelte, Git, Mathematica, Blender, high-performance computing (HPC), C++, Fusion360, SQL, PostgreSQL, Grafana.
- **Leadership & Collaboration:** Multi-disciplinary collaboration leadership, scientific communication (6+ international conferences, multiple first-author publications), technical documentation.

SOFTWARE DEVELOPMENT

- ltempy / GitHub - Tools for the analysis, simulation, and presentation of Lorentz TEM data.
- frctl / GitHub - A single-page web app for exploring the Mandelbrot and Burning Ship fractals.

SELECTED PUBLICATIONS

- Hybrid Skyrmions in Magnetic Multilayer Thin Films are Half-Integer Hopfions. **W.S. Parker**, J. Reddinger, B.J. McMorran. *Phys. Rev. B.* **2024.** DOI: 10.1103/PhysRevB.110.224420.
- Real Space Imaging of Hybrid Skyrmion Textures in Magnetic Multilayer Thin Films. **W.S. Parker**, J. Reddinger, R. Moraski, E.E. Fullerton, S.A. Montoya, B.J. McMorran. *Phys. Rev. B.* **2025.** DOI: 10.1103/zlzp-chdg
- Combining Lorentz TEM and SEM with Polarization Analysis to Uncover Fractional Topological Spin Textures in Fe/Gd Multilayer Thin Films. **W.S. Parker**, S.A. Montoya, E.E. Fullerton, B.J. McMorran. *Microscopy and Microanalysis.* **2024.** DOI: 10.1093/mam/ozae044.507.
- Exact Design of Complex Amplitude Holograms for Producing Arbitrary Scalar Fields. C.W. Johnson, J.S. Pierce, R.C. Moraski, A.E. Turner, A.T. Greenberg, **W.S. Parker**, B.J. McMorran. *Opt. Express.* **2020.** DOI: 10.1364/OE.393224.

AWARDS, GRANTS, SCHOLARSHIPS

- Best Student Presentation – Finalist, Magnetism and Magnetic Materials. 2023
- Honorable Mention, National Science Foundation Graduate Research Fellowship. 2020
- Student Scholar Award, Microscopy and Microanalysis. 2019
- Graduate First Year Fellowship, University of Oregon. 2018-2019
- OMQ Director's Fellowship, University of Oregon. 2018

RELEVANT COURSEWORK

- **Graduate (University of Oregon):** Quantum Information, Quantum Optics and Laser Physics, Solid State Physics, Space Physics, Scientific Computation

LINKS

wsparker.com | LinkedIn | GitHub | Google Scholar | Curriculum Vitae